CtrlPay — Sandbox Execution Evidence

2026 Shenzhen FinTechathon · International Track · Topic A Submission Checklist Item 06 — Execution evidence

This report is generated by `tests/generate-execution-evidence.test.ts` against the real engine, Zustand sandbox store, proposal service and hash-chained audit logger. The complete machine-readable log is in `docs/EXECUTION_EVIDENCE_SANDBOX_LOGS.json`.

Field	Value
Tenant (userId)	`student_usm_001`
Sandbox start clock	`2026-03-12T09:00:00.000Z`
Final sandbox clock	`2026-03-31T23:59:59.999Z`
Audit entries	11
Audit chain verified	YES
Final ledger	balance RM 7521.5 · defenseVault RM 16.4 · rewardPool RM 22.4

The 8-step judge workflow

Step 1 — Default-OFF safety gate (T0 advisory only)

- Sandbox clock: `2026-03-12T09:00:00.000Z`
- Ledger after this step: balance RM 7540 · defenseVault RM 0 · rewardPool RM 24


```
{
  "masterAutonomyEnabled": false,
  "t0": {
    "shortfallProbability": 0,
    "cfar95": 0,
    "es95": 0,
    "intraMonthLiquidityBreachProb": 0,
    "lowestProjectedBalanceRM": 6802.24,
    "riskLevel": "HIGH"
  },
  "t3Attempt": {
    "ok": false,
    "reason": "MASTER_AUTONOMY_OFF: CtrlPay is in T0 advisory/analysis mode only."
  }
}
```


Step 2 — Payment interceptor — Shopee RM 180 impulse buy

- Sandbox clock: `2026-03-12T09:00:00.000Z`
- Ledger after this step: balance RM 7540 · defenseVault RM 0 · rewardPool RM 24

```
{
  "seed": 42,
  "shortfallProbability": 0,
  "cfar95": 0,
  "es95": 0,
  "intraMonthLiquidityBreachProb": 0,
  "lowestProjectedBalanceRM": 6802.24,
  "stratifiedShortfallProbability": 0,
  "projectedMonthEndBalance": 6544.97,
  "projectedMonthEndBalanceWithIncomeRM": 7529.87,
  "dynamicDailyAllowance": 332.82,
  "riskLevel": "HIGH",
  "limitBreach": {
    "weekly": true,
    "monthly": true
  },
  "studentUnits": {
    "mixedRiceMeals": 20,
    "bubbleTea": 12
  }
}
```


Step 3 — 24h Commitment Quarantine lock (early claim refused)

- Sandbox clock: `2026-03-12T09:00:00.000Z`
- Ledger after this step: balance RM 7540 · defenseVault RM 0 · rewardPool RM 24


```
{
  "proposalId": "prop_quarantine_0001",
  "authorizationRef": "USER_CLICK_APPROVE_PROPOSAL_0001",
  "terms": {
    "kind": "QUARANTINE",
    "targetItem": "Shopee impulse buy (RM 180)",
    "itemPrice": 180,
    "premiumReward": 9,
    "quarantineHours": 24,
    "shortfallProbability": 0,
    "cfar95": 0,
    "riskLevel": "HIGH"
  },
  "status": "LOCKED_24H",
  "earlyClaimRefused": true,
  "rewardPoolUnchangedAt": 24
}
```


Step 4 — Fast-forward +24h and claim the quarantine reward

- Sandbox clock: `2026-03-13T09:00:00.000Z`
- Ledger after this step: balance RM 7540 · defenseVault RM 9 · rewardPool RM 15

```
{
  "proposalId": "prop_quarantine_0001",
  "status": "CLAIMED",
  "rewardClaimed": 9,
  "rewardPoolBefore": 24,
  "rewardPoolAfter": 15,
  "defenseVaultBefore": 0,
  "defenseVaultAfter": 9
}
```


Step 5 — Zombie Radar and Spotify student downgrade sweep

- Sandbox clock: `2026-03-13T09:00:00.000Z`
- Ledger after this step: balance RM 7540 · defenseVault RM 9 · rewardPool RM 15


```
{
  "zombieTags": {
    "netflix_my": "ZOMBIE",
    "spotify_my": "ACTIVE",
    "youtube_my": "ENDANGERED",
    "disney_plus_my": "SUSPECTED_ZOMBIE",
    "icloud_my": "ACTIVE"
  },
  "studentDowngradeCandidates": [
    "spotify_my",
    "youtube_my"
  ],
  "studentVerified": true,
  "spotifyFullPriceRM": 15.9,
  "spotifyStudentPriceRM": 8.5,
  "monthlySavings": 7.4,
  "sweepStatus": "PENDING_NEXT_BILLING_VERIFICATION",
  "defenseVaultNotCreditedYet": 9
}
```


Step 6 — Jump to Month-End — two-stage sweep settlement

- Sandbox clock: `2026-03-31T23:59:59.999Z`
- Ledger after this step: balance RM 7540 · defenseVault RM 16.4 · rewardPool RM 22.4

```
{
  "sweepStatus": "VERIFIED",
  "monthlySavings": 7.4,
  "defenseVaultGrowth": 7.4,
  "rewardPoolGrowth": 7.4,
  "conservationHolds": true,
  "dailySummaryFired": true,
  "monthEndRoast": {
    "wastedMonthly": 109.9,
    "wastedAnnual": 1318.8,
    "units": {
      "mixedRiceMeals": 12,
      "bubbleTea": 7
    },
    "unitsAnnual": {
      "mixedRiceMeals": 146,
      "bubbleTea": 87
    },
    "tone": "GENTLE"
  }
}
```


Step 7 — Back-Tap quick logging syncs the dynamic daily budget

- Sandbox clock: `2026-03-31T23:59:59.999Z`
- Ledger after this step: balance RM 7521.5 · defenseVault RM 16.4 · rewardPool RM 22.4


```
{
  "sandboxClock": "2026-03-31T23:59:59.999Z",
  "trigger": "iOS Back-Tap / double-tap B / sandbox button → single quick-log path",
  "amountRM": 18.5,
  "balanceBefore": 7540,
  "balanceAfter": 7521.5,
  "dynamicDailyAllowanceBefore": 6980.1,
  "dynamicDailyAllowanceAfter": 6961.6,
  "authorizationRef": "USER_AUTHORIZED_BACKTAP_TRANSFER"
}
```


Step 8 — Cryptographic audit-chain verification

- Sandbox clock: `2026-03-31T23:59:59.999Z`
- Ledger after this step: balance RM 7521.5 · defenseVault RM 16.4 · rewardPool RM 22.4


```
{
  "verified": true,
  "entryCount": 11,
  "genesisPrevHash": "GENESIS",
  "finalEntryHash": "761782c8007ffaea7ff84d88bdaf67c07e68b865e683cdae5645d36a452b6afe"
}
```

Sample [SANDBOX] operation-log lines

```
2026-03-12T09:00:00.000Z  [SANDBOX] simulatePaymentIntent Shopee 180 HIGH
2026-03-12T09:00:00.000Z  [SANDBOX] setMasterAutonomy true
2026-03-12T09:00:00.000Z  [SANDBOX] setTier T3 true
2026-03-12T09:00:00.000Z  [SANDBOX] applyDecision APPROVED prop_quarantine_0001
2026-03-12T09:00:00.000Z  [SANDBOX] claim refused (not ready) prop_quarantine_0001
2026-03-13T09:00:00.000Z  [SANDBOX] advanceClock 1773306000000 → 1773392400000 2 event(s)
2026-03-13T09:00:00.000Z  [SANDBOX] claimQuarantineReward prop_quarantine_0001 9
2026-03-13T09:00:00.000Z  [SANDBOX] setStudentVerified true
2026-03-13T09:00:00.000Z  [SANDBOX] applyDecision APPROVED prop_sweep_0001
2026-03-31T23:59:59.999Z  [SANDBOX] advanceClock 1773392400000 → 1775001599999 21 event(s)
2026-03-31T23:59:59.999Z  [SANDBOX] quickLogTransaction EXPENSE Tealive & Campus Lunch (Back-
Tap) 18.5
```

Final SHA-256 audit-chain digest

- Genesis `prevHash`: `GENESIS`
- Final `entryHash`: `761782c8007ffaea7ff84d88bdaf67c07e68b865e683cdae5645d36a452b6afe`
- `verifyChain(userId)`: **true**

The chain is `entryHash = SHA-256(canonicalJson(entry))` with `prevHash` binding each entry to its predecessor, so altering any earlier entry breaks every later hash.

Ledger conservation

`defenseVault` growth from the Spotify sweep equals the verified `monthlySavings` (RM 7.4), and the same amount funds the `rewardPool`. Vault growth never exceeds verified savings, and the rewardPool is funded only by verified sweep savings.

Sandbox/demo numbers only. No real money moves; `defenseVault` is a non-interest-bearing ledger bucket.